The problem of reproducibility

Metabolic coupling between soil aerobic methanotrophs and denitrifiers in rice paddy fields

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We nead a real title: Subject and predicate, our thesis. The problem of reproducibility: Publication of papers with mislabelled data, incomplete information, faulty figures, incongruent conclussions.

1 Abstract

2 Introduction

In 2024 about 523.8 million metric tons of rice were consumed worldwide. Besides, over 50 percent of the world population depends on rice for about 80 percent of its food requirements. To meet this demand, a total of 200 million km^2 are destined to cultivating rice in paddy fields, more than 10% of the cropland worldwide. This huge buisness has it's downside, consuming large amounts of N-fertilizer and generating vast quantities of methane (CH_4), a greenhouse gas with 30 times more warming potential than CO_2. For those reasons, understanding the biological processes produced in these environments, is a key requirement for mantaining the production of an essensial food while reducing its environmental cost.

Rice paddy fields are a very interesting ecosystem, containig, in the rainy months, a layer of water that creates an oxygenation gradient, perfect for the development of a huge variety of microorganisms.

One of the main problems of actual science is the problem of reproducibility.

We will reproduce some of the results of this study, reassuring stadistic results and commenting possible flaws.

3 Methods

Reproduction of methods and results

Experiment reproduced: Use of Galaxy for data processing and QIIME2 analysis. Use of Python and Bash for genus classification and nirS and nirK analysis.

Statystical replication with R and figures.

4 Results

Not sufficient data
Stupid p-values
Failure of analysis

5 Discussion

I hate this paper

6 References